Maddox Nehls

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Education

Oregon State University **Corvallis, OR**

Honors Bachelor of Science in Computer Science GPA: 3.96

Experience

Full Stack Engineer Mar 2026 – Present

AVIBE, Beaverton, OR *Full-time*

- Own production web apps for multiple clients end to end, from database schema and APIs to UI, shipping staff portals, enrollment systems, reporting tools and admin dashboards.
- Maintain and modernize long-running ColdFusion systems alongside new Laravel and Vue codebases.

Software Engineer Intern Apr 2025 – Aug 2025

Ally, Charlotte, NC *Internship*

- Shipped customer-facing features for Ally's digital banking apps in React, TypeScript, Java and Spring Boot.
- Resolved production bugs and security vulnerabilities in a highly regulated financial codebase, in Agile sprints.

Technical Solutions Analyst Apr 2024 – Sep 2024

Intel, Portland, OR *Internship*

- Traced compatibility failures across Intel processor platforms to root cause and turned performance data into fixes.
- Trained sales and other non-technical teams on how Intel hardware and software fit together, in plain language.

Software Developer Intern Nov 2023 – Feb 2024

MissionNext, Remote *Internship*

- Built backend features for a job-matching platform in PHP and Laravel, from database models to API endpoints.
- Profiled and rewrote slow MySQL and PostgreSQL queries, and fixed user-reported bugs alongside the support team.

Full Stack Engineer Intern Sep 2022 – Sep 2023

CareerVillage.org, Remote *Internship*

- Shipped user-requested features for a nonprofit student career platform with React, Node.js and Python.
- Analyzed engagement data in SQL to help set priorities, and wrote guides for students breaking into software engineering.

Projects

Kiln | *Rust, AVX2 SIMD, axum* ***GitHub***

- Built an LLM inference engine from scratch in Rust with no ML framework: tokenizer, transformer forward pass, KV cache and sampler, with output matching Hugging Face transformers token for token.
- Wrote AVX2 kernels and int8/int4 quantization to reach 346 tokens/s on a CPU, on par with llama.cpp (350), and served the model behind an OpenAI-compatible HTTP API.

Tandem | *TypeScript, React, WebSockets, PostgreSQL* ***GitHub***

- Built a local-first collaborative notes app on a CRDT sync engine written from scratch. It applies a real 259,778-edit editing trace 8.4× faster than Yjs (194 ms vs 1.6 s) at about 1 byte per edit.
- Added live cursors, share links, offline merging on reconnect and version history, with Playwright end-to-end tests.

Tracewright | *Go, PostgreSQL, Next.js, Python* ***Live Demo***

- Built a flight recorder for AI agents that records every model and tool call, lines up two runs step by step to show where they diverge, and replays a run from any step with a fix.
- Wrote a Go ingest service that sustained 24,400 events/s at 25 ms p99 using bounded queues with 429 backpressure and batched Postgres COPY, plus a CI check that fails pull requests when agent quality drops.

Undercut | *TypeScript, Playwright, PostgreSQL, Next.js* ***GitHub***

- Built a price tracker for sites that block bots: a collector on a home connection runs a real browser, and the scheduler learns which sites need it from each site's cloud block rate.
- Leased jobs exactly once through Postgres (FOR UPDATE SKIP LOCKED with lease expiry) and required two consecutive checks to agree before alerting, so flickering prices never send false alerts.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, Go, Rust, PHP, C/C++, C#, SQL, ColdFusion, Kotlin, Ruby, Swift
Frontend: React, Next.js, Vue, Angular, React Native, Redux, Zustand, Tailwind, HTML, CSS, jQuery, Bootstrap
Backend & Data: Node.js, Laravel, Spring Boot, ASP.NET, FastAPI, Django, Flask, GraphQL, WebSockets, PostgreSQL, pgvector, MySQL, MongoDB, Prisma, Supabase
Tools: Git, GitHub Actions, Docker, Playwright, AWS, Azure DevOps, Jenkins, Vercel, Figma